

3.2 Fundamentals of data structures

3.2.1 Data structures and abstract data types

3.2.1.1 Data structures

Content	Additional information
Be familiar with the concept of data structures.	It may be helpful to set the concept of a data structure in various contexts that students may already be familiar with. It may also be helpful to suggest/demonstrate how data structures could be used in a practical setting.

3.2.1.2 Single- and multi-dimensional arrays (or equivalent)

Content	Additional information
Use arrays (or equivalent) in the design of solutions to simple problems.	A one-dimensional array is a useful way of representing a vector. A two-dimensional array is a useful way of representing a matrix. More generally, an n -dimensional array is a set of elements with the same data type that are indexed by a tuple of n integers, where a tuple is an ordered list of elements.

3.2.1.3 Fields, records and files

Content	Additional information
Be able to read/write from/to a text file.	
Be able to read/write data from/to a binary (non-text) file.	